

MJ2246 - Mission to the Moon

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1 Introduction

1.1 Mission Requirements

1. Estimated $\Delta v = 3.5$ km/s
2. Travel time $\tau = 208$ days
3. Travel from GTO ($h_{pE} = 620$ km; $h_{aE} = 36000$ km) to Moon polar operational orbit (between 300 km and 3000 km in altitude over Moon surface)
4. Initial Mass: $m_{S/C} = 400$ kg

1.2 Assumptions

1. Propellants Evaluated: Xenon, Krypton, Hydrogen, Argon
2. Tank Performance Factor, $K : 10^5 m^2/s^2$,
3. Tank Temperature at Filling, $T : 20^\circ C$
4. Specific Power for Processing Unit (PPU), γ_{PPU} : 150 W/kg,
5. Specific Power for Electric Thrusters, γ_{Thrust} : 250 W/kg
6. PPU Efficiency, η_{PPU} : 90%
7. Thruster Efficiency, η_{Thrust} : 45 %
8. "Non-impulse dependent system mass factor", x : 5 %

2 Method

2.1 Power Generator

Based on the long mission time and space operating conditions, there are two efficient solutions for generating energy - nuclear fission and solar energy. The nuclear power provides much lower efficiency when comparing to solar arrays [1] [2]. The propulsion requires high efficiency, and the choice of solar arrays is justified this way. Some of the solar array data is available from Reference [3]. Assessed power generators with their efficiencies are listed in Table 1.

Table 1: Potential electric power generators

Type	Name	γ_{PowGen} (W/kg)	ρ_{PowGen} (W/m ²)
MMRTG	NASA in-house	2.8	-
RPS	10kWe Kilopower System (NASA STMD)	6.5	-
Solar array	Upgraded X4/X5 flexible type (TJ)	136	267
Solar array	Recommended - flexible (TJ)	125	250

MMRTG - Multi-mission Radioisotope Thermoelectric Generator; *RPS* - Radioisotope Power Systems

The array technology selected is the upgraded X4/X5 flexible type with Triple-Junction cells. The parameter that has been taken into account to make the choice is the specific power γ_{PowGen} . The reason is that the overall specific power γ and thus the design of the propulsion system and its performance are affected by this parameter, as can be seen later in the next paragraphs. Since, the higher the specific power the better the performance, the power generator with the highest specific power is chosen, accounting for 136 W/kg.

2.2 Propellant Evaluation

The analysis is based on the initial parameters and the Power Generator specs mentioned, respectively in Section 1.1, 1.2 and 2.1. The evaluated propellants are Xenon, Krypton, Hydrogen, and Argon, representing the most common options for which electric thrusters are developed. Their specific parameters in terms of molecular mass M and gas compressibility factor z used to perform the analysis are listed in Table 2. Molar masses can be found from literature, and as compressibility factor (assuming ideal gas) can be calculated as follows:

$$Z = \frac{p}{\rho * R_{spec} * T} \quad (1)$$

where p is pressure, ρ is gas density, R_{spec} is specific gas constant and T is temperature.

Table 2: Propellant specific parameters

	Xenon	Krypton	Hydrogen	Argon
M (kg/kmol)	131.3	82.9	2.01	39.9
z	0.3	0.8	1	0.99

2.2.1 Analysis Step 1

The way the propellant evaluation is performed is shown below (formulas based on Reference [4] and [5]):

- Calculation of the overall Specific Energy, γ

$$\gamma = \left(\frac{1}{\gamma_{PowGen}} + \frac{1}{\gamma_{Thrust}} + \frac{1}{\gamma_{PPU}} \right)^{-1} \quad (2)$$

- Calculation of the overall power conversion efficiency, η

$$\eta = \prod_i \eta_i = \eta_{PPU} \cdot \eta_{Thrust} \quad (3)$$

- Evaluation of the optimum thruster exhaust velocity v_{e-opt} for all the propellants
- Calculation of the required propellant mass m_p and thrust level F required based on the mission requirements:

$$m_p = m_{S/C} \left(1 - e^{-\frac{\Delta v}{v_{e-opt}}} \right), \quad (4)$$

$$F = \frac{m_p v_{e-opt}}{\tau}. \quad (5)$$

This analysis allows to evaluate the best option in terms of the propellant that can be chosen to fulfill the mission requirements in the most efficient way. Since the goal is to maximise the I_{ssp} , the choice is based on the one that provides the best performance in that direction.

2.3 Thruster Evaluation

A pre-selection of the possible optimal thrusters is done in this step. This selection is based on the following criteria:

- Thrust Level evaluated in the previous step, based on the mission requirements
- Thruster Efficiency, η_{Thrust} since it affects the overall efficiency and so the amount of power required from the power generator
- Thruster Specific Power γ_{Thrust} , since it affects the overall Specific Power.
- Specific Impulse I_{sp} , since it affects the amount of propellant required

After selecting the thrusters that match the thrust level required by the mission requirements and the propellant chosen, the focus is in particular on the efficiency first and the specific power then, because they should be the parameters that affect the most the final mass. The thrusters selected with their specifications are listed in Table 3.

Table 3: Thruster datasheet

Number of Thrusters	1	1	2	3
Name of the Thruster	PPS 1350	NSTAR	R-800	ST40
Efficiency	50.3%	62%	55.4%	40%
Mass (kg)	5.3	8.33	1.5(x2)	1.1 (x3)
I_{sp} (s)	1730	3120	1550	1850
Power Required by the Thruster (W)	1500	2290	900 (x2)	600 (x3)
Average Thrust (N)	0.089	0.092	0.053 (x2)	0.028 (x3)
Specific Power (W/kg)	283	275	600	545.45

2.3.1 Analysis Step 2

The analysis performed for the evaluation of the thrusters is summarised below:

- Re-calculation of γ and η by using Equations 2 and 3, considering the thruster specific γ_{Thrust} and η_{Thrust} from the datasheet
- Evaluation of the optimum thruster exhaust velocity v_{e-opt} and propulsion system mass fraction m_{PS}/m_{SC} for $\Delta v = 3500$ m/s
- Calculation of the propulsion system mass m_{ps} :

$$m_{PS} = massfraction * m_{SC} \quad (6)$$

- Estimation of the mass of propellant required m_p using Equation 4, and tank volume and mass:

$$V_{tank} = \frac{m_p}{\rho_{Xe}}, \quad m_{tank} = m_p \frac{(1 - \rho_{tank})}{\rho_{tank}}, \quad (7)$$

where the ρ_{Xe} is based on the existing Ti-6Al-4V Xenon tank for space applications developed by MT-Aerospace, S-XTA. In this model, the Xenon is kept at 187 bar and it has a density of 1833 kg/m^3 [6]. The other parameter ρ_{tank} is the propellant mass to full tank ratio and it has a typical value of 0.92 for the modern Xenon tank as can be estimated by the mentioned tank and the literature [7].

- Calculation of the power required from the power generator to operate the thruster, P_{PowGen} :

$$P_{PowGen} = \frac{F v_{e-opt}}{2\eta} \quad (8)$$

where F is evaluated with Equation 5 and v_{e-opt} .

- Evaluation of mass and size of the power generator:

$$m_{PowGen} = \frac{P_{PowGen}}{\gamma_{PowGen}}, \quad S_{PowGen} = \frac{P_{PowGen}}{\rho_{PowGen}} \quad (9)$$

- Feasibility Analysis of the analysed thrusters.

2.3.2 Feasibility Analysis

The feasibility of the configuration chosen is checked based on the following criteria:

- The calculated thruster power P must not exceed the one provided in the datasheet for that specific thruster
- The calculated I_{sp} must not exceed the real one listed in the datasheet

The check is performed with the following steps:

- Power Check:

$$P = P_{PowGen} \eta \quad (10)$$

- Specific Impulse check:

$$I_{sp} = \frac{v_{e-opt}}{g} \quad (11)$$

If both the parameters don't exceed the respective ones provided in the datasheet, then the configuration is feasible as it has been evaluated in the previous steps. If just one requirement is not satisfied, then the corresponding one provided by the supplier is considered a limiting factor for the exit velocity evaluation and a v_{e-max} is estimated from it. If both the requirements are not filled, then the minimum exit velocity obtained from them is taken into account to evaluate a new feasible configuration for the thruster analysed. This part of the feasibility evaluation is explained below:

- Evaluation maximum exit velocity:

$$v_{e-max} = \frac{2P_{datasheet}}{F}, \quad v_{e-max} = I_{sp_{datasheet}}g \quad (12)$$

- Evaluation of the new propulsion system mass fraction
- Repetition of all the steps mentioned in Section 2.3.1 from the propulsion system mass calculation to the evaluation of the size and mass of the power generator
- New Power evaluation
- Thrust Level Check with Equation 5 and the v_{e-max} evaluated: it has to match with the thrust level calculated in the thruster evaluation.
- Check if the power generator mass exceeds the maximum feasible one provided by the calculated propulsion system mass.

3 Results and Discussion

3.1 Propellant analysis

The results in terms of optimum thruster exhaust velocity v_{e-opt} , system specific impulse I_{ssp} , propellant mass m_p and thrust required F , obtained for each propellant are listed in the table below 4. These values are obtained taking into account $\gamma = 55.491$ W/kg and $\eta = 0.405$. Xenon has the highest I_{ssp} and therefore is chosen as the propellant for our mission.

	Table 4: Propellant evaluation results			
	$v_{e-opt}(m/s)$	$I_{ssp}(Ns/kg)$	$m_p(kg)$	$F(N)$
Xenon	29203.3	13172.4	45.1784	0.0734
Krypton	31395.7	12177.3	41.9511	0.0774
Hydrogen	102996.1	3921.6	13.3644	0.00583
Argon	36231.4	10749.3	36.8328	0.074258

Note: The results have been obtained by considering $x = 5\%$

3.2 Thruster Analysis

The thruster evaluation performed as shown in Section 2.3 yields the results listed in the table 5. These results allow to discuss the criteria taken into account for the thruster pre-selection. Indeed, the best performance in terms of I_{ssp} , m_{ps} and m_p are obtained with the NSTAR that is characterised

Table 5: Thruster comparison data

Number of Thrusters	1	1	2	3
Name of the Thruster	PPS 1350	NSTAR	R-800	ST40
Propulsion Systems Mass (kg)	108	84	108	108
Propellant Mass (kg)	74.5418	43.24	82.2437	70.18
Tank Volume (m^3)	0.0407	0.0236	0.0449	0.0383
Tank Mass (kg)	6.482	3.76	7.152	6.103
Power Required from the Power Generator (W)	3051.73	2031.38	2266.3	1850.42
Power Required for the Thruster (W)	1381.52	1133.51	1129.98	666.15
Average Thrust (N)	0.070394	0.0736	0.069587	0.0708
Payload Mass (kg)	270	301	275	278
Total Efficiency	0.4527	0.558	0.4986	0.36
Overall Specific Power (W/kg)	56.97	56.63	63.75	63.08
Exit Velocity (m/s)	16971.3	30596	15205.5	18142
Isp (Ns/kg)	11828.7	15500	11510.6	11842.8
Isp (s)	1730	3120	1550	1850
Solar Panels Mass (kg)	22.4392	14.9366	16.6639	13.6060
Feasibility Mass of Solar Panels (kg)	16.5334	24.6700	11.9614	25.4741

by the highest I_{sp} and η_{Thrust} . Regarding the γ_{Thrust} , it slightly affects the overall γ and so the performance, even though there is a large difference in terms of it between the thrusters, as can be seen between R-800 and ST40, and PPS 1350 and NSTAR. The importance of the I_{sp} can be noticed by analysing the R-800. Despite it has the second highest efficiency and the highest specific mass, its I_{sp} is the lowest between the thrusters analysed and it results in the worst option in terms of I_{ssp} , m_{ps} and m_p . The data listed in Table 5 are obtained after the feasibility check. As can be noticed, the I_{sp} performance are the same of the ones provided by the datasheet, while the powers are lower. The reason is because the specific impulse has been the limiting factor for all of them: during the thruster evaluation the first results have provided higher I_{sp} , exceeding the feasible requirements. It means that all the exit velocities shown are not the optimal ones but the maximum attainable to meet the requirements of the mission.

It is important to notice that during the mass analysis, according to equation:

$$m_{PowGen_feasi} = \frac{m_{PS_feasi}}{1.05} - (m_{tank_mass} + m_{P_feasi} + m_{Thruster}) \quad (13)$$

the feasible mass of the Power Generator, reported in Table 5 calculated at the end of the feasibility analysis resulted lower for the PPS-1350 and the R-800 thrusters, meaning that, in order to achieve the desired thrust level much more power should be needed than the one can actually be produced with the other constraints we have applied regarding I_{ssp} etc. This resulted in the exclusion of the said thrusters from the choices of optimal thruster for the mission.

3.2.1 Optimal Solution

The optimal configuration for the mission analysed is found in the NSTAR thruster. Since the goal of the mission is to choose the option that maximise the I_{ssp} , this option attains the best performance in it. It directly affects the mass of the propulsion system that is the lowest between the four configurations, allowing to have an higher payload capacity. Furthermore, the high performance in

terms of efficiency and I_{sp} have also a positive contribution, since they allow to have both an lower propulsion system mass, propellant mass required and a smaller power generator compared to the amount of power required by the thruster to operate.

4 Conclusion

This study provides a comparative evaluation of the different propellants, power generators and thrusters in order to achieve the mission requirements. There are two main power generator types for long missions available on market - nuclear fission and solar arrays, from which solar arrays show the highest efficiency. From all the propellants analyzed, Xenon allows to obtain the highest system-specific impulse I_{ssp} . As noticed from the results obtained, the parameters characterising each thruster affect its performance in different ways. It can be stated that I_{sp} has a big impact on the final propulsion system mass and I_{ssp} , since the propellant mass is strictly dependent on it. Efficiency is a crucial factor in propulsion system choice, as it helps to minimize propulsion system mass and its effects can be more or less relevant depending on the power generator specifications.

References

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